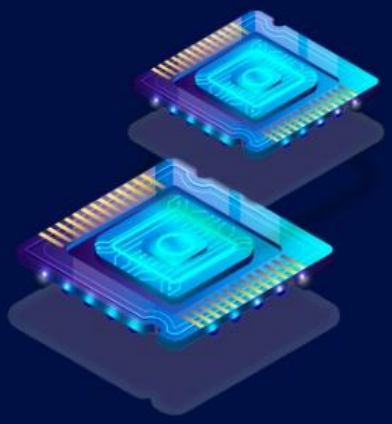
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Embedded Systems & Robotics with IoT

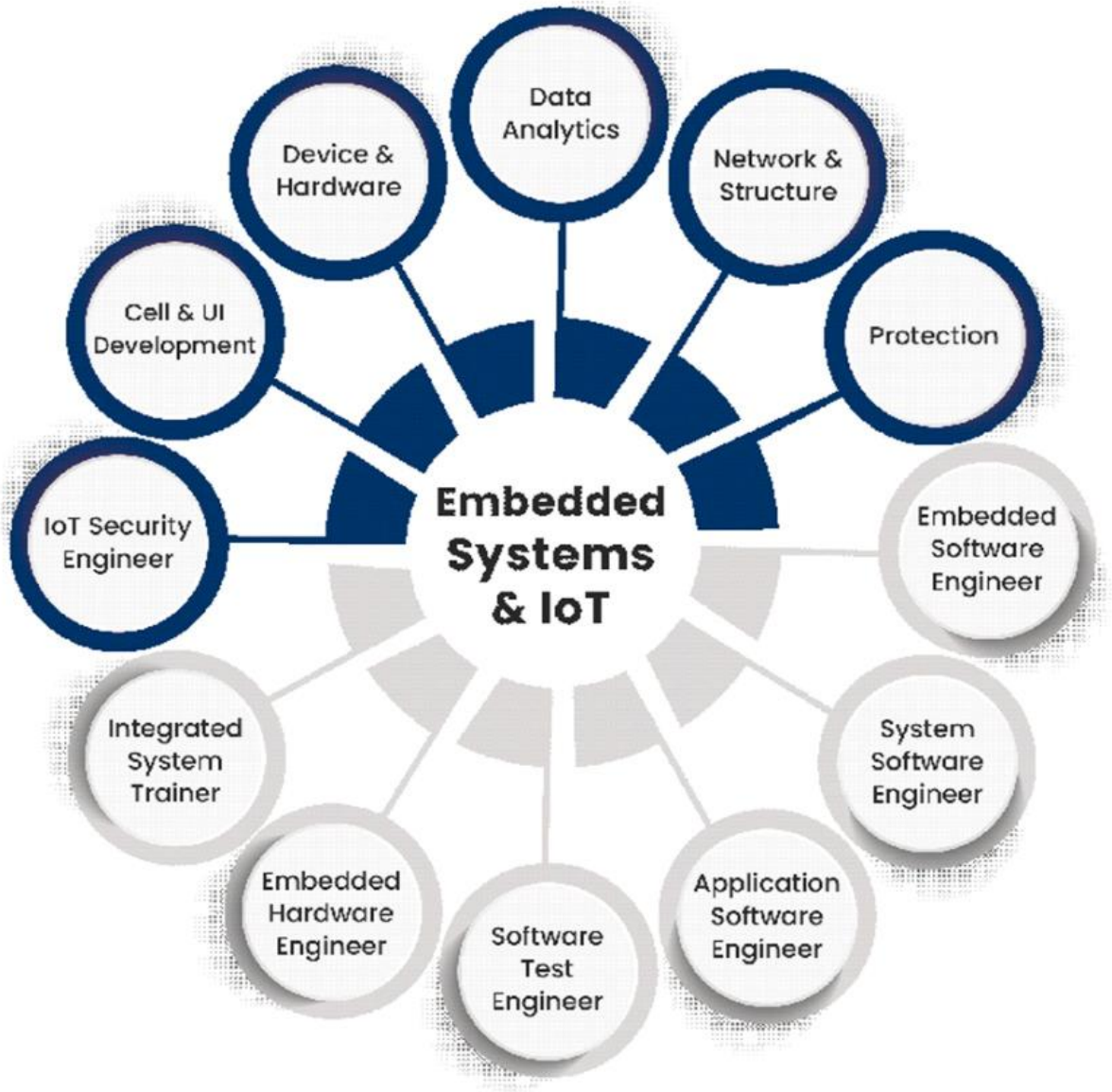


Become proficient in AVR Microcontroller | Arduino | Embedded C++ | Display | Sensors | Relay | Motors| Switches | Communication | Bluetooth | RFID | GSM | IoT



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Carrer Opportunities in Embedded Systems & Robotics with IoT



Top Skills Needed

for high paying Career in Embedded Systems & IoT

- Data Security
- Cloud Computing
- Business Intelligence
- Networks
- IoT Hardware
- Application Design
- Mobile Applications

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A Brief About **UPFLAIRS**

UpFlairs is an innovative **educational technology** company with a clear mission to elevate the skills and employability of students throughout India.

Our dedicated team is committed to fostering the next generation of tech talent, equipping them with cutting-edge skills in emerging technologies and has educated more than 47K students all over the globe including IITs, NITs, Deemed Universities and other colleges.

We offer the courses that are most trending technologies of the recent era in the fields like **AI/ML, Data Science, Cloud Computing, DevOps, Full Stack Web Development, Embedded Systems, IoT, and Robotics.** These courses are meticulously designed to provide students with the practical skills and knowledge required to excel in tech-driven careers, making them not just job-ready but industry leaders of tomorrow.

We are not only limited to training the youth of the country but also provide Lab setups to various Colleges and Universities and Project solutions to other companies for the domains like **AI-ML, IOT , ROBOTICS AND CLOUD**.



Who Should Go for This Embedded Systems & Robotics with IoT ?



College Student

Start shaping your career right from college time by learning in-demand skills like Embedded Systems & Robotics with IoT.



Graduate

Time for you to upskill, prepare yourself for a bright future, and kickstart a career in Embedded Systems & Robotics with IoT



College Dropout/Job Seeker

Not sure which career is right for you ? Not Finding any jobs? Learn Embedded Systems & Robotics with IoT with us to land your first job.



Freelancer

Whether you are an existing freelancer or want to become one, IoT is the most- rewarding skill for freelancers.



Looking to Switch Career

Not Satisfied with your current job profile? Switch to Embedded Systems & Robotics with IoT, an easy-to-learn, most impressive, and highly-paid skill.

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Why Learn Embedded Systems & Robotics with IoT with Upflairs Only?

Trusted by Learners

Our training and support system intrigue learners.

Expert Trainers

Professional Trainers having experience of delivering more than 30k students. We ensure the high quality training always aimed at career building.

Completely Practical-Oriented

We offer 100% practical training with regular assessment, assignment and projects.

Industry-Recognized Certification

The certificate received by the learners on course completion is valid nationally and internationally. You can easily share , add it to your resume and explore great opportunities.

Online & Offline Batches

You can upskill yourself from anywhere by joining our live training batches Also we provide classroom training in Jaipur.

Career Assistance

We help you build a great career as a developer. Our team offers you career guidance, pepares you for interview, and assists you in finding job.

Hands-On Projects

You will get to work on multiple projects during the training so that you can gain some experience.

Comprehensive Curriculum

The course includes most comprehensive Curriculum, covering all breadths and depths in detail.

Project Titles

Final Project (1)

"Voice Controlled IoT based System for Controlling & Monitoring Indoor Farm"

Projects Major + Minor (31+)

1. *Timer Based Alarm*
2. *Traffic Light Prototype*
3. *Message Display Screen*
4. *Smart Countdown Bulb*
5. *Arithmetic Calculator*
6. *SA RE GA MA Generator*
7. *Smart Door Lock*
8. *Simple Virtual Messenger*
9. *Digital Scale*
10. *Color Sensing Device*
11. *Visitor Counter*
12. *Smart Street Lights*
13. *Torch Operated Bulb*
14. *Smart Home Fan/Cooler*
15. *Clap Controlled Switch*
16. *Intrusion Detector for Theft Protection*
17. *Automatic Water Level Indicator*
18. *Smart Water Pump*
19. *Digital Nose for Fire/Smoke*
20. *Automatic Hand Sanitizer*
21. *Parking Assistance*
22. *Digital Water Level Measurement Tool*
23. *Line Follower Robotic Vehicle*
24. *Obstacle Avoider/Follower Robotic Vehicle*
25. *Grid Solving Robotic Vehicle (Mite)*
26. *Smart Home Automation System*
27. *Office Attendance System*
28. *Fast Tag based Toll Collection System*
29. *Metro Train Ticketing System"*
30. *Live Location Tracker for Vehicle Accident Detection*
31. *Wireless Indoor Sensor Data Logger for Centralized Controlling*

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Embedded Systems & Robotics with IoT

Part 1 (Duration – 30 Days/4 Weeks)

Week 1 (15 Hours)	**Introduction** (2.5 Hours)	• IoT Definition & Emergence • Real World applications of IoT • Architecture; Technology Stack; • Hardware and Software Platforms; • Role of Embedded Systems in IoT • Features of AVR Mega328/2560 based Arduino • Description of Complete Hardware Structure & Pin-out • Introduction with Embedded C++ • Programming an Arduino Board • **Practical 1: Creating first Arduino Sketch**
	GPIO (2.5 Hours)	• Configuring I/O and controlling I/O • Setting mode of pin as input and output • Driving pin high and low • LED Interfacing • **Practical 2: Interfacing Simple LED** • **Practical 3: Reading Input Pin Data**
	GPIO – Display (7 Segment) (2.5 Hours)	• Need, Types & Working Concept • Interfacing of SSD with Microcontroller • Multiplexing for Multiple Digits • **Practical 5: Generating 0 to 9 Counter** • **Practical 6: Generating 00 to 99 and 000 to 999 Counter**
	GPIO (1 Hours)	• Working of Relay & Buzzer • **Practical 7: Controlling Relay as a Switch using Arduino** • **Practical 8: Controlling Buzzer with Arduino**
	Project (1.5 Hours)	• ***Project 1: "Timer Based Alarm"***
	Project (2.5 Hours)	• ***Project 2: "Traffic Light Prototype"***
	Liquid Crystal Display (2.5 Hours)	• Introduction to 16×2 LCD • LCD programming with Arduino • **Practical 9: Printing Simple String, Numbers on LCD** • **Practical 10: Scrolling message Display** • Writing order change
Week 2 (15 Hours)	**Liquid Crystal Display** (1 Hours)	• Custom character on LCD • ***Practical 11: Generating Custom Characters on LCD***
	Project (1.5 Hours)	• ***Project 3: "Message Display Screen"***
	Keys/Switches (1.5 Hours)	• Concept of Pullup and Pulldown • The types of switches • ***Practical 12: Interfacing Switches with MCU***
	Project (1 Hours)	• ***Project 4: "Smart Countdown Bulb"***
	Keypad Matrix	• Working of Keypad Matrix

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	(2.5 Hours)	• ***Practical 13: Working of Keypad Matrix***
	Project (2.5 Hours)	• ***Project 5: "Arithmetic Calculator"***
	PWM/Timer (1.5 Hours)	• PWM in Arduino • ***Practical 14: PWM Generation with Arduino*** • Timer Configuration for Tone Generation • ***Practical 15: Generating Tones using Arduino*** • Counter Usage for Time Calculation • ***Practical 16: Calculating Time Elapsed***
	Project (1 Hours)	• ***Project 6: "SA RE GA MA Generator"***
	Motors (1.5 Hours)	• DC Motor: Working, Motor Driver Circuit; H-Bridge Concept • ***Practical 17: Controlling Simple DC Motor Direction using Serial*** • Servo Motor: Servo Mechanism, Working • ***Practical 18: Controlling Servo Motor with Arduino***
	Project (1 Hours)	• ***Project 7: "Smart Door Lock"***
Week 3 (15 Hours)	**Bluetooth: Serial Communication** (1.5 Hours)	• Serial Communication with Arduino • **Bluetooth**: Protocol, Working, Interfacing • ***Practical 19: Sending Data Serially*** • ***Practical 20: Receiving Data Serially***
	Project (1 Hours)	• ***Project 8: "Simple Virtual Messenger"***
	Sensors – Part 1 (0.5 Hours)	• Using inbuilt ADC of Arduino • Voltage Comparator IC for Digital Mode • **IR Sensor** – Concept, Working Circuit, Interfacing • Line of Sight Phenomenon • IR Range, IR Sensor Circuit • Voltage Comparator IC for Digital Mode • ***Practical 21: Reading IR Sensor Variation***
	Project (2 Hours)	• ***Project 9: "Digital Scale"*** • ***Project 10: "Color Sensing Device"*** • ***Project 11: "Visitor Counter"*** • ***Project 12: "Smart Street Lights"***
	Sensors – Part 2 (1.5 Hours)	• LDR Sensors: Definition, Phenomenon • ***Practical 22: Taking Analog Light Intensity Values*** • Temperature Sensor: Construction, Concept, Scaling Factor • ***Practical 23: Fetching Real-time temperature data***
	Project (1 Hour)	• ***Project 13: "Torch Operated Bulb"*** • ***Project 14: "Smart Home Fan/Cooler"***
	Sensors – Part 3 (2.5 Hours)	• **SOUND SENSOR:** Concept of Sound Detection, Working Circuitry • ***Practical 24: Controlling LED with Sound Sensor*** • **PIR SENSOR:** Concept of Pyro-Electricity, Calibration • ***Practical 25: Detecting Human Presence with PIR*** • **Moisture Sensor:** Construction, Concept, Working Circuit

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		• ***Practical 26: Interfacing Moisture Sensor*** • **Gas Sensor:** LPG/Smoke Detector, Construction, Concept, Pin-out • ***Practical 27: Interfacing Gas Sensor***
	Project (2.5 Hours)	• ***Project 15: "Clap Controlled Switch"*** • ***Project 16: "Intrusion Detector for Theft Protection"*** • ***Project 17: "Automatic Water Level Indicator"*** • ***Project 18: "Smart Water Pump"*** • ***Project 19: "Digital Nose for Fire/Smoke"***
	Sensors – Part 4 (1 Hours)	• Ultrasonic Sensor: Concept of Ultrasonic Waves • Working of HC-SR04 • ***Practical 28: Interfacing of Ultrasonic Sensor with Arduino*** • ***Practical 29: Measuring & Displaying Distance of Objects***
	Project (1.5 Hours)	• ***Project 20: "Automatic Hand Sanitizer"*** • ***Project 21: "Parking Assistance"*** • ***Project 22: "Digital Water Level Measurement Tool"***
Week 4 (15 Hours)	**Interrupts** (2.5 Hours)	• Definition, Significance, Advantages • Various Interrupts (Hardware, Software) • Interrupt Service Routines • ***Practical 30: Hardware Interrupt Simulation*** • ***Practical 31: Real-time control of DC Motors***
	Robotics (1 Hours)	• Dynamics of Robotics • Sensor Bots • ***Practical 32: Assembling Robotic Structure***
	Robotics (4 Hours)	• ***Project 23: "Line Follower Robotic Vehicle"*** • ***Project 24: "Obstacle Avoider/Follower Robotic Vehicle"*** • ***Project 25: "Grid Solving Robotic Vehicle (Mite)"***
	PCB Designing & Fabrication (5 Hours)	• Tools, Rules for PCB Designing • Tracing & Fabrication • ***Practical 33: PCB Fabrication***
	Project (2.5 Hours)	• ***Project 26: "Smart Home Automation System"***
Part 2 (Duration – 15 Days/3 Weeks)		
Week 5 (15 Hours)	**RFID** (1 Hours)	• RFID: Technology, Concept, Circuit • Interfacing of RFID Reader with AVR • ***Practical 34: Displaying Tag Information and Matching***
	Project (1.5 Hours)	• ***Project 27: "Office Attendance System"*** • ***Project 28: "Fast Tag based Toll Collection System"*** • ***Project 29: "Metro Train Ticketing System""***
	GSM (2.5 Hours)	• Introduction to mobile communication • Introduction to GSM terminology • GSM Frequency bands, SIM

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		• GSM AT Commands • ***Practical 35: GSM Network testing in GSM Modem*** • ***Practical 36: GSM AT Commands testing for reading SMS, dial a number*** • ***Practical 37: GSM call answering / halt operations*** • ***Practical 38: Sending SMS from GSM modem to another mobile***
	GPS (2.5 Hour)	• Introduction to GPS • Satellites and its Orbits • Range, Accuracy and Tracking • ***Practical 39: GPS Network Testing & Location Coordinates***
	Project (2.5 Hours)	• ***Project 30: "Live Location Tracker for Vehicle Accident Detection"***
	Wi-Fi (2.5 Hour)	• **ESP8266**: Pinout., configuration, working, interfacing • ESP AT Commands • ***Practical 40: ESP as a server configuration***
	Project (2.5 Hour)	• ***Project 31: "Wireless Indoor Sensor Data Logger for Centralized Controlling"***
Week 6 (15 Hours)	**IoT Platform** (5 Hour)	• Basic Concepts of Networking (Types, Topologies, IP, HTTP, URL) • Various Data Handling Requests in HTTP • What is IoT Platform? • What is API? • ***Practical 41: ThingSpeak Platform Channels*** • ***Practical 42: ThingSpeak Platform ThingHTTP*** • ***Practical 43: ThingSpeak Platform Reactions***
	MQTT - IoT Platform (2.5 Hour)	• Understanding Message Queueing Telemetry for IoT • The concept of Subscribers and Publishers • Setting up MQTT on Python • ***Practical 44: MQTT based private chat room***
	MQTT - IoT Platform (2.5 Hour)	• Adafruit IoT Platform for MQTT • IFTTT Platform for Google-Assistant and other Controls • ***Practical 45: MQTT based Switching***
	Final Project (5 Hour)	• ***"Voice Controlled IoT based System for Controlling & Monitoring Indoor Farm"***

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